

Beginner's Guide

Different Types of Keys in Databases and Their Uses



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Primary Key

It's like a unique ID for each row in a table.

**It ensures that each row is unique and
cannot be null.**



2

Foreign Key

A foreign key links a table to another table by referencing its primary key, creating relationships.

Example: An "Orders" table might connect to "Customers" using a foreign key referencing the "Customer ID."



3

Composite Key

When a single column isn't enough to uniquely identify a record, we combine two or more columns.

Example: An "Invoice" table might use "Invoice Number" and "Invoice Date" together to uniquely identify each invoice.



Super Key

Any group of columns in a super key can uniquely identify a record, but only one becomes the primary key.

Example: A "Student" table might have "Student ID," "Last Name," and "First Name" as a super key, but only "Student ID" might be chosen.



5

Candidate Key

Like a contender for primary key, a candidate key is unique but might not be the chosen one.

Example: In the "Student" table, both "Student ID" and "Email" (if unique) could be candidate keys.



6

Unique Key

Similar to a primary key, a unique key ensures no duplicates, but allows one record to be missing a value.

Example: A "Product" table might have a unique key on "Product Code" to allow for a product without a code yet.



7

Alternate Key

This is a backup option for unique identification within a table, offering flexibility.

Example: A "User" table might have a primary key on "User ID" and an alternate key on "Email" for login purposes.



8

Natural Key

This is a key that naturally identifies a row.

Example: An email address



Surrogate Key

When there's no natural unique identifier,
we create an artificial key.

Example: An "Employee" table might use
an auto-generated "Employee ID" as a
surrogate key.



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Secondary Key

These don't identify records directly, but act like helpful indexes to speed up searches.

Example: A "Products" table might have a secondary key on "Category" for faster searches by category.

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